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PUBLICATION LIST

Under review or in preparation

1. **V. Strokov**. “The Glow of Eternal Black Holes”. Under review, *Class. Quant. Grav.* (in revision after the first round). 2026. arXiv: [2608.05270 \[gr-qc\]](#).
2. **V. Strokov**, A. Kankani, and S. T. McWilliams. “Multi-source Fisher-matrix analysis of overlapping sources in LISA: I. Massive black hole binaries”. In preparation; provisional title.
3. **V. Strokov** and S. T. McWilliams. “Multi-source Fisher-matrix analysis of overlapping sources in LISA: II. Galactic double white dwarfs”. In preparation; provisional title.
4. LISA Synthetic UCB Catalogs Project (incl. **V. Strokov**). “LISA Synthetic UCB Catalogs Project: Effect of Astrophysical Uncertainties on the Predicted Galactic DWD Population”. In preparation.
5. LISA Synthetic UCB Catalogs Project (incl. **V. Strokov**). “LISA Synthetic UCB Catalogs Project: Code Comparison”. In preparation.

Refereed

1. N. Geissler, **V. Strokov**, C. Kümmeler, S. Kushnarev, and E. Berti. “Hankel low-rank matrix approximation for gravitational-wave data analysis”. In: *Phys. Rev. D* 114.6 (2026), p. 064079. DOI: [10.1103/j9j3-fn3q](#). arXiv: [2603.15740 \[astro-ph.HE\]](#).
2. R. Ebadi, **V. Strokov**, E. H. Tanin, E. Berti, and R. L. Walsworth. “LISA double white dwarf binaries as Galactic accelerometers”. In: *Phys. Rev. D* 111.4 (2025), p. 044023. DOI: [10.1103/PhysRevD.111.044023](#). arXiv: [2405.13109 \[gr-qc\]](#).
3. K. Kritos, **V. Strokov**, V. Baibhav, and E. Berti. “Dynamical formation of black hole binaries in dense star clusters: Rapid cluster evolution code”. In: *Phys. Rev. D* 110.4 (2024), p. 043023. DOI: [10.1103/PhysRevD.110.043023](#). arXiv: [2210.10055 \[astro-ph.HE\]](#).
4. L. Reali, R. Cotesta, A. Antonelli, K. Kritos, **V. Strokov**, and E. Berti. “Intermediate-mass black hole binary parameter estimation with next-generation ground-based detector networks”. In: *Phys. Rev. D* 110.10 (2024), p. 103002. DOI: [10.1103/PhysRevD.110.103002](#). arXiv: [2406.01687 \[gr-qc\]](#).
5. **V. Strokov** and E. Berti. “Quasimonochromatic LISA sources in the frequency domain”. In: *Phys. Rev. D* 109.10 (2024), p. 104013. DOI: [10.1103/PhysRevD.109.104013](#). arXiv: [2312.00121 \[gr-qc\]](#).
6. **V. Strokov**, G. Fragione, and E. Berti. “LISA constraints on an intermediate-mass black hole in the Galactic Centre”. In: *Mon. Not. Roy. Astron. Soc.* 524.2 (2023), pp. 2033–2041. DOI: [10.1093/mnras/stad2002](#). arXiv: [2303.00015 \[astro-ph.HE\]](#).
7. **V. Strokov**, G. Fragione, K. W. K. Wong, T. Helfer, and E. Berti. “Hunting for intermediate-mass black holes with LISA binary radial velocity measurements”. In: *Phys. Rev. D* 105.12 (2022), p. 124048. DOI: [10.1103/PhysRevD.105.124048](#). arXiv: [2109.08154 \[astro-ph.HE\]](#).
8. A. A. Popov, **V. N. Strokov**, and A. A. Surdyayev. “A proof-of-concept neural network for inferring parameters of a black hole from partial interferometric images of its shadow”. In: *Astronomy and Computing* 36 (2021), p. 100467. DOI: [10.1016/j.ascom.2021.100467](#). arXiv: [2311.16227 \[astro-ph.HE\]](#).

9. **V. N. Strokov** and Sh. Khlghatyan. “The orbital Lense–Thirring precession in a strong field”. In: *General Relativity and Gravitation* 51.7 (2019). DOI: [10.1007/s10714-019-2563-9](https://doi.org/10.1007/s10714-019-2563-9). arXiv: [1906.05309](https://arxiv.org/abs/1906.05309) [gr-qc].
10. **V. N. Strokov**, V. N. Lukash, and E. V. Mikheeva. “Black-and-white hole as a space-time with integrable singularity”. In: *International Journal of Modern Physics A* 31 (2016), p. 1641018. DOI: [10.1142/S0217751X16410189](https://doi.org/10.1142/S0217751X16410189).
11. T. R. P. Caramês, M. H. Daouda, J. C. Fabris, A. M. Oliveira, O. F. Piattella, and **V. Strokov**. “The Brans-Dicke-Rastall theory”. In: *European Physical Journal C* 74 (2014), p. 3145. DOI: [10.1140/epjc/s10052-014-3145-3](https://doi.org/10.1140/epjc/s10052-014-3145-3). arXiv: [1409.2322](https://arxiv.org/abs/1409.2322) [gr-qc].
12. V. N. Lukash, E. V. Mikheeva, and **V. N. Strokov**. “The early universe and cosmogenesis”. In: *Physics of Particles and Nuclei* 44 (2013), pp. 415–421. DOI: [10.1134/S1063779613030064](https://doi.org/10.1134/S1063779613030064).
13. V. N. Lukash and **V. N. Strokov**. “Space-Times with Integrable Singularity: Black-White Holes and Astrogenic Universes”. In: *International Journal of Modern Physics A* 28 (2013), p. 1350007. DOI: [10.1142/S0217751X13500073](https://doi.org/10.1142/S0217751X13500073). arXiv: [1301.5544](https://arxiv.org/abs/1301.5544) [gr-qc].
14. S. V. Repin, V. N. Lukash, and **V. N. Strokov**. “Analytical approximation for the Fe K α emission line in quasar spectra”. In: *Astronomy Reports* 52 (2008), pp. 1–11. DOI: [10.1134/S1063772908010010](https://doi.org/10.1134/S1063772908010010). arXiv: [0705.2574](https://arxiv.org/abs/0705.2574) [astro-ph].
15. **V. Strokov**. “On convergence to equilibrium in strongly coupled Bogoliubov’s oscillator model”. In: *Infinite Dimensional Analysis, Quantum Probability, and Related Topics* 10.4 (2007), pp. 573–589. DOI: [10.1142/S0219025707002865](https://doi.org/10.1142/S0219025707002865). arXiv: [quant-ph/0612202](https://arxiv.org/abs/quant-ph/0612202).
16. **V. N. Strokov**. “Cosmological status of Lagrangian theory of density perturbations”. In: *Astronomy Reports* 51 (2007), pp. 431–434. DOI: [10.1134/S1063772907060017](https://doi.org/10.1134/S1063772907060017). arXiv: [astro-ph/0612397](https://arxiv.org/abs/astro-ph/0612397).

Non-Refereed

1. **V. N. Strokov**, A. A. Popov, and Sh. G. Khlghatyan. “Numeric integration of geodesic equations in Schwarzschild’s field in different coordinate frames”. In: *Proceedings of the 59th MIPT science conference. General and applied physics*. Ed. by V.V. Kiselev. In Russian. Moscow Institute of Physics and Technology. Moscow: MIPT, 2016, pp. 75–77.
2. V. N. Lukash, E. V. Mikheeva, and **V. N. Strokov**. “From the Cosmological Model to the generation of the Hubble flow”. In: *Physics Uspekhi* 55 (2012), pp. 204–209. DOI: [10.3367/UFNe.0182.201202k.0216](https://doi.org/10.3367/UFNe.0182.201202k.0216). arXiv: [1110.6820](https://arxiv.org/abs/1110.6820) [astro-ph.CO].
3. V. N. Lukash, E. V. Mikheeva, and **V. N. Strokov**. “Generation of cosmological flows in general relativity”. In: *Physics Uspekhi* 55.8 (2012), pp. 831–837. DOI: [10.3367/UFNe.0182.201208k.0894](https://doi.org/10.3367/UFNe.0182.201208k.0894). arXiv: [1211.6619](https://arxiv.org/abs/1211.6619) [gr-qc].
4. **V. Strokov**. “ADM Formulation of Cosmological Perturbation Theory”. In: *Twelfth Marcel Grossmann Meeting on General Relativity*. 2012, p. 1382.
5. V. N. Lukash and **V. N. Strokov**. “Geometries with integrable singularity – black/white holes and astrogenic universes”. In: *ArXiv e-prints* (2011). arXiv: [1109.2796](https://arxiv.org/abs/1109.2796) [astro-ph.CO].
6. V. N. Lukash and **V. N. Strokov**. *The sources of geometries with an integrable singularity: black/white holes and astrogenic universes*. Tech. rep. 3. In Russian. Lebedev Physical Institute, 2011.
7. **V. N. Strokov**. *A quantum model of the quasi-Friedmannian universe*. Tech. rep. 7. In Russian. Lebedev Physical Institute, 2011.
8. **V. Strokov**. “Equivalence of two approaches in theory of cosmological density perturbations”. In: *Nuovo Cimento B* 122 (2007), pp. 1399–1404.

OPEN-SOURCE SOFTWARE

- Confusion-foreground pipeline of the LISA Synthetic UCB Catalogs project (author of the pipeline)
github.com/Synthetic-UCB-Catalogs/Galaxy-scripts (under **Foregrounds/**)
- Hankel low-rank matrix approximation for GW data (in collaboration with N. Geissler)
github.com/CosmoVlad/lisa-hankel
- Neural-network recognition of black-hole shadows (in collaboration with A. A. Popov and A. A. Surdyaev)
github.com/CosmoVlad/silhouette
- Appearance of eternal black holes, with an interactive diagram viewer
github.com/CosmoVlad/white-hole-shadow
- Sonification of LISA signals
github.com/CosmoVlad/gw-sounds